Ericsson PE Node WHI-PE-14 Routing Protocol Flap Troubleshooting SOP

Node: WHI-PE-14 | **Vendor:** Ericsson | **Severity:** Minor | **Alarm Code:** ALARM_6577

Description

The Routing Protocol Flap on Ericsson PE Node WHI-PE-14 at Whitefield East POP is a complex issue that can arise from a multitude of factors, including BGP route churn, OSPF adjacency loss, and ISIS LSP flooding. This Standard Operating Procedure (SOP) is designed to troubleshoot and resolve this failure domain on the specified node. The PE node is a critical component of the network infrastructure, responsible for routing traffic between different network segments. The Ericsson equipment, being a multi-protocol device, supports a variety of routing protocols, including BGP, OSPF, and ISIS. The Routing Protocol Flap can occur due to various reasons such as misconfiguration, hardware issues, or software bugs. It is essential to troubleshoot this issue promptly to prevent network downtime and ensure optimal network performance. The diagnostic process will involve analyzing the routing protocol logs, checking the node's configuration, and verifying the network connectivity. The resolution steps will focus on correcting the root cause of the issue, which may involve reconfiguring the node, restarting the routing protocols, or replacing faulty hardware.

Pre-requisite Checks

- description: Verify that the Whitefield East POP is accessible and that the remote hands are dispatched to the site.

- description: Check the redundant link status to ensure that it is operational and not contributing to the Routing Protocol Flap.

- description: Backup the node's configuration before proceeding with any troubleshooting or resolution steps.

Potential Root Causes

Cause	Probability
BGP Route Churn	High
OSPF Adjacency Loss	Medium
ISIS LSP Flooding	Low

Diagnostic Commands & Expected Outputs

```
Command: show bgp summary, Expected Output: BGP table version is 123, main routing table version is 123., Description: This command displays the BGP summary, including the number of peers, routes, and updates. It helps to identify if there are any issues with the BGP routing protocol.
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```
Command: show ospf neighbor, Expected Output: OSPF neighboring routers: 2, Description: This command displays the OSPF neighboring routers, including their IDs, state, and timer information. It helps to identify if there are any issues with the OSPF routing protocol.

Command: show isis database, Expected Output: ISIS database: 10 LSP(s) up to 0x12345678, Description: This command displays the ISIS database, including the number of LSPs and their contents. It helps to identify if there are any issues with the ISIS routing protocol.
```

Resolution Steps

Step 1: Step: Reconfigure the BGP node to remove the problematic route, Command: configure replace bgp route 123.123.123.123, Description: This command removes the problematic BGP route from the routing table, resolving the BGP Route Churn issue.

Step 2: Step: Restart the OSPF routing protocol to re-establish adjacencies, Command: configure restart ospf, Description: This command restarts the OSPF routing protocol, re-establishing adjacencies between the OSPF neighboring routers and resolving the OSPF Adjacency Loss issue.

Step 3: Step: Clear the ISIS LSP flooding by reconfiguring the ISIS node, Command: configure replace isis lsp 123.123.123.123, Description: This command clears the ISIS LSP flooding by reconfiguring the ISIS node, resolving the ISIS LSP Flooding issue.

Post-Check Verification

```
Command: show bgp summary, Expected Output: BGP table version is 123, main routing table version is 123., Description: Verify that the BGP summary shows no issues and the routing table is stable.

Command: show ospf neighbor, Expected Output: OSPF neighboring routers: 2, Description: Verify that the OSPF neighboring routers show no issues and the adjacencies are re-established.

Command: show isis database, Expected Output: ISIS database: 10 LSP(s) up to 0x12345678, Description: Verify that the ISIS database shows no issues and the LSPs are stable.
```

Roll-back Procedures

- Step: Restore the node's configuration to the previous state, Command: configure restore, Description: Restore the node's configuration to the previous state in case the fix fails or introduces new issues.

- Step: Restart the routing protocols to re-establish connections, Command: configure restart all, Description: Restart the routing protocols to re-establish connections and restore the network to its previous state.